



STM32F3 Technical Training

For reference only

Refer to the latest documents for details



Reset and Clock control RCC

- Reset:
 - Initialize the device
 - Wakeup device
 - Safety functions (watchdog)
- Clocks:
 - Select appropriate clock source:
 - Internal
 - External
 - Select appropriate speed:
 - High speed
 - Low speed
 - Speed regulation
 - Modify clock parameters for:
 - Core
 - Peripherals
 - Security functions:
 - In case of clock source malfunction

- System RESET

- Resets all registers except some RCC registers and Backup domain
- Sources:
 - Low level on the NRST pin (External Reset)
 - WWDG & IWWDG end of count condition
 - A software reset (through NVIC)
 - Low power management reset
 - Option byte loader reset (FORCE_OBL bit)

- Power RESET

- Resets all registers except the Backup domain
- Sources:
 - Power On/Power down Reset (POR/PDR)
 - Exit from STANDBY

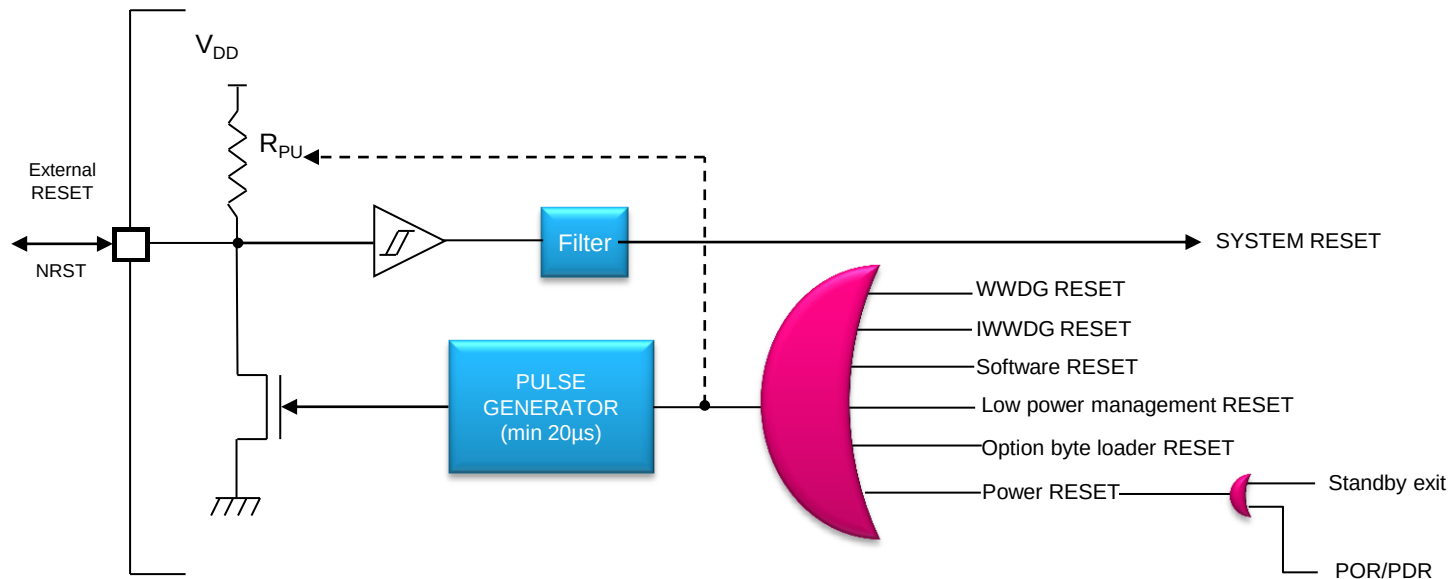
- Backup domain RESET

- Resets in the Backup domain: RTC registers + Backup Registers + RCC_BDCR register
- Sources:
 - BDRST bit in RCC_BDCR register
 - POWER Reset

Reset block diagram

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- Reset sources in STM32F3 family and their relation to RESET pin:



Clock features (1/2)

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- System Clock (SYSCLK) sources:
 - HSE (High Speed External oscillator or crystal)
 - 4MHz to 32MHz,
 - can be bypassed by user clock
 - HSI (High Speed Internal RC):
 - factory trimmed internal RC oscillator 8MHz +/- 1%
 - PLL x2, x3, .. x16
 - From HSE or HSI/2
 - 16MHz – 72MHz output
- Additional clock sources:
 - LSI (Low Speed Internal RC):
 - ~40kHz internal RC
 - LSE (Low Speed External oscillator):
 - 32.768kHz
 - can be bypassed by user clock
 - Configurable driving strength (power/robustness compromise)

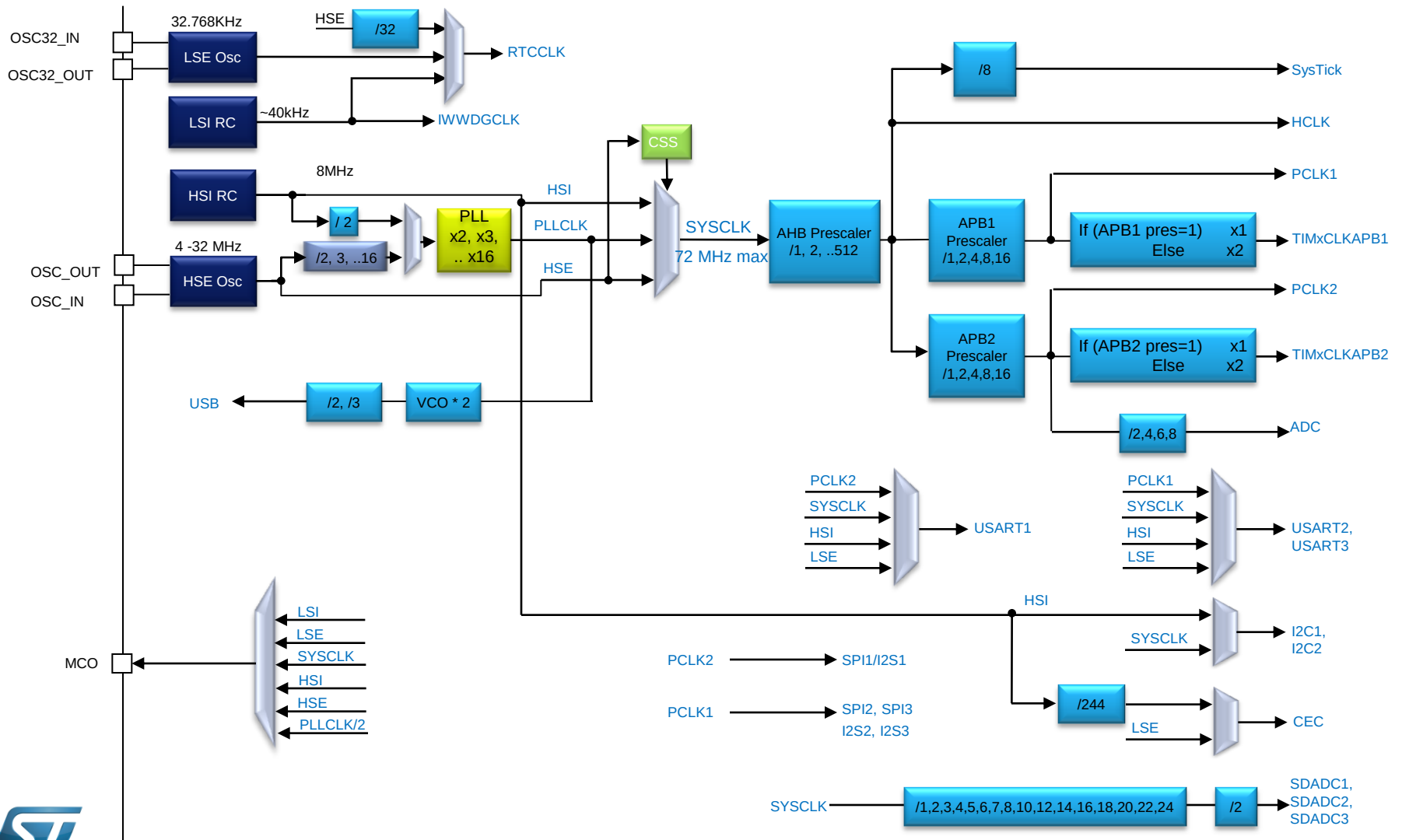
Clock features (2/2)

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- Clock-out capability on the MCO:
 - LSI, LSE, SYSCLK, HSI, HSE, PLL/2
- Clock Security System (CSS) to switch to backup clock:
 - In case of HSE clock failure
 - Enabled by SW w/ interrupt capability linked to NMI
 - Could generate BREAK for Timers
- RTC Clock sources:
 - LSE, LSI and HSE/32
- USART, I2C & CEC have multiple possible clock sources:
 - Possibility to wakeup device if there is no system clock:
 - For USART: HSI, LSE
 - For I2C: HSI
 - For HDMI-CEC: LSE, HSI

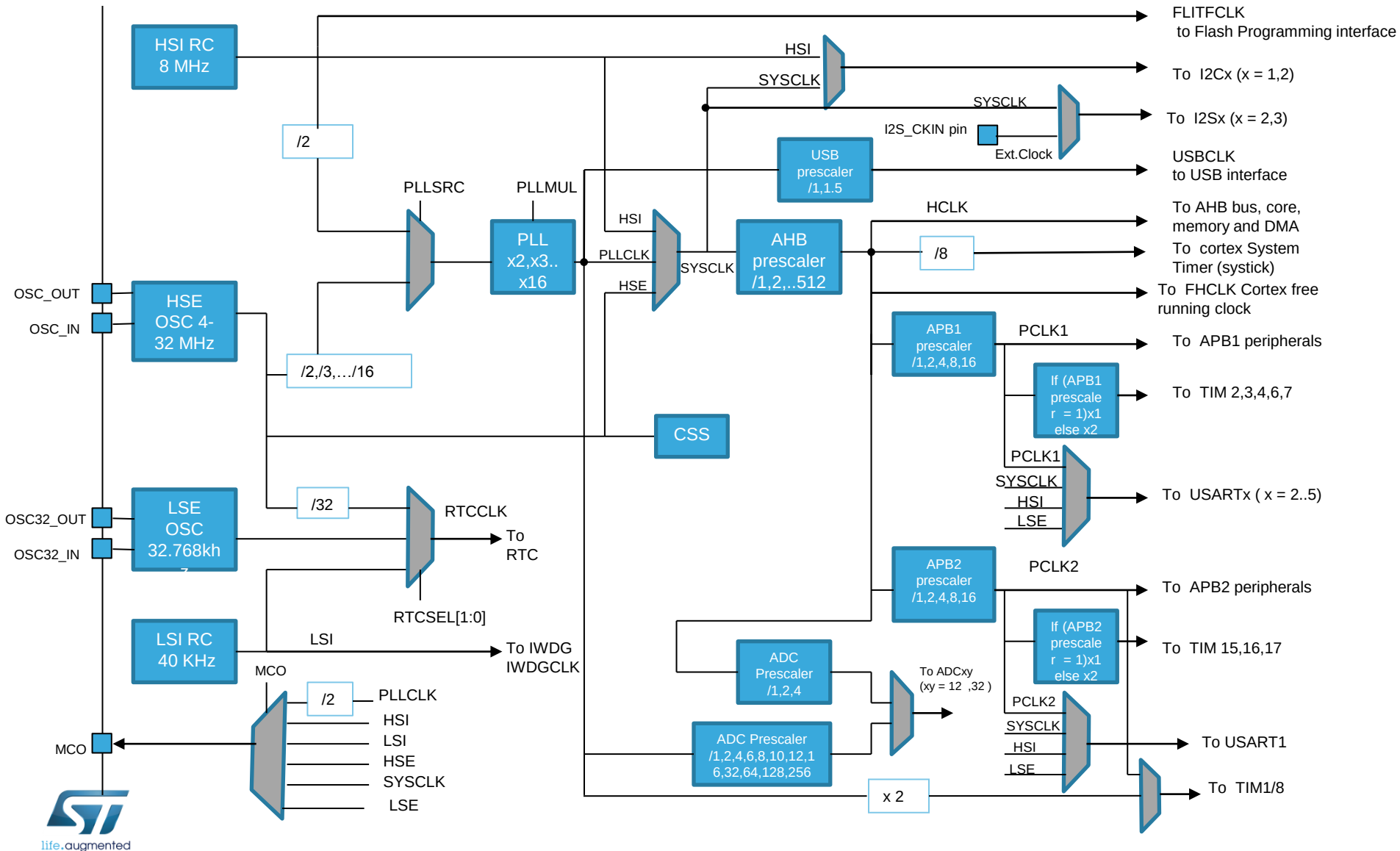
Clock scheme STM32F37x

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Clock scheme STM32F30x

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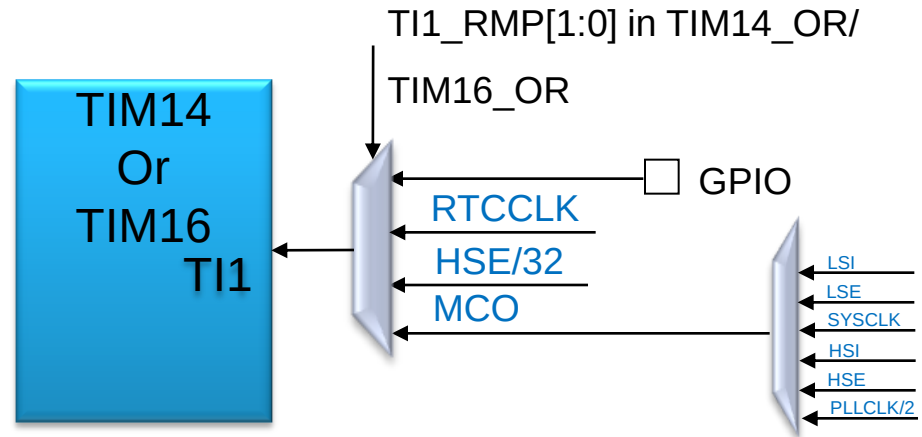


HSI/LSI/ext. clock measurement

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- TIM14 (in F37x) and TIM16 (in F30x) input capture can be triggered by:

- GPIO pin
- RTCCLK
- HSE/32
- MCO output



- Purposes:

- **Measure HSI frequency** using the precise LSE clock. HSI is used as system clock. Knowing the (more precise) LSE frequency we can determine the HSI frequency.
- **Measure the LSI frequency** using HSE or HSI. To fine tune IWWDG and/or RTC timing (if LSI used as RTC clock).
- Have rough indication of the **frequency of external crystal** – by comparing HSI and HSE/32

- What is the maximum AHB and APB1 and APB2 clock frequencies ?
- What is the purpose of connecting LSE clock to TIM14/16 CH1 input capture and how it could be done?
- What is the purpose of the CSS?



Thank you !

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